Capital university

Faculty of Engineering

Computer and Systems Engineering Department

Task L03

Submitted by

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3rd Computer and Systems

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Q3)

D-

Q3] d]

Path	Queue
	340 [A]
340 [A]	355 [A,S] 425 [A,Z] 428 [A,T]
[A,S]	425 [A,Z] 428 [A,T] 366 [A,S,R] [A,S,F]
[A,S,R]	366 [A,S,R,C] 425 [A,Z] 428 [A,T] 409 [A,S,F]

[A,S,R,C] Goal

path A → S → R → C

cost = 366

```
def path_f_cost(path):
    g_cost = 0
    for (node, cost) in path:
        g_cost += cost

    last_node = path[-1][0]
    h_cost = h_table[last_node]

    f_cost = g_cost + h_cost
    return f_cost, last_node


def a_star_search(graph, start, goal):
    visited = []
    queue = [[(start, 0)]]

    while queue:
        queue.sort(key=path_f_cost)
        path = queue.pop(0)
        node = path[-1][0]

        if node in visited:
            continue

        visited.append(node)

        if node == goal:
            return path
        else:
            adjacent_nodes = graph.get(node, [])
            for (node2, cost) in adjacent_nodes:
                new_path = path.copy()
                new_path.append((node2, cost))
                queue.append(new_path)
```

lab:

Q4)

```
***     expanded_states:         ['A', 'B', 'C', 'D']
*** PASS: test_cases\q4\graph_manypaths.test
***     solution:                ['1:A->C', '0:C->D', '1:D->F', '0:F->G']
***     expanded_states:         ['A', 'B1', 'C', 'B2', 'D', 'E1', 'F', 'E2']

### Question q4: 3/3 ###


Finished at 14:18:55

Provisional grades
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Question q4: 3/3
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Total: 3/3

Your grades are NOT yet registered.  To register your grades, make sure
to follow your instructor's guidelines to receive credit on your project.

PS C:\Users\User\Downloads\search>
```

Q5)

```
***     pacman layout:           tinyCorner
***     solution length:                 28

### Question q5: 3/3 ###


Finished at 14:26:58

Provisional grades
==================
Question q2: 3/3
Question q5: 3/3
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Total: 6/6
```

Q6)

```
'North', 'North', 'East', 'East', 'South', 'South', 'South', 'South', 'South', 'North
 'East', 'East', 'North', 'North']
path length: 106
*** PASS: Heuristic resulted in expansion of 1139 nodes

### Question q6: 3/3 ###


Finished at 14:30:58

Provisional grades
==================
Question q4: 3/3
Question q6: 3/3
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Total: 6/6

Your grades are NOT yet registered.  To register your grades, make sure
to follow your instructor's guidelines to receive credit on your project.
```